TECHJAM 2026

SciOdyssey.

Long-Horizon Autonomous ML Research Agent

Autonomous by default.
Supervisable by design.

ROBUST · SUPERVISABLE · BROAD SEARCH

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01 | PROBLEM · FAILURE MODES

Why current agents fail

Read the task, inspect data, engineer features, train and evaluate. Reproduce the baseline, then improve it.

01 Closed-world workflow

Predefined tools and recovery paths struggle with unexpected failures.

Design response: an open coding-agent harness that can inspect and repair.

02 Single trajectory

A long-lived context carries prior reasoning into the next experiment.

Design response: fresh Scientist context with independent META review.

03 Tree search

Repeated context reconstruction can cost more than the code edit itself.

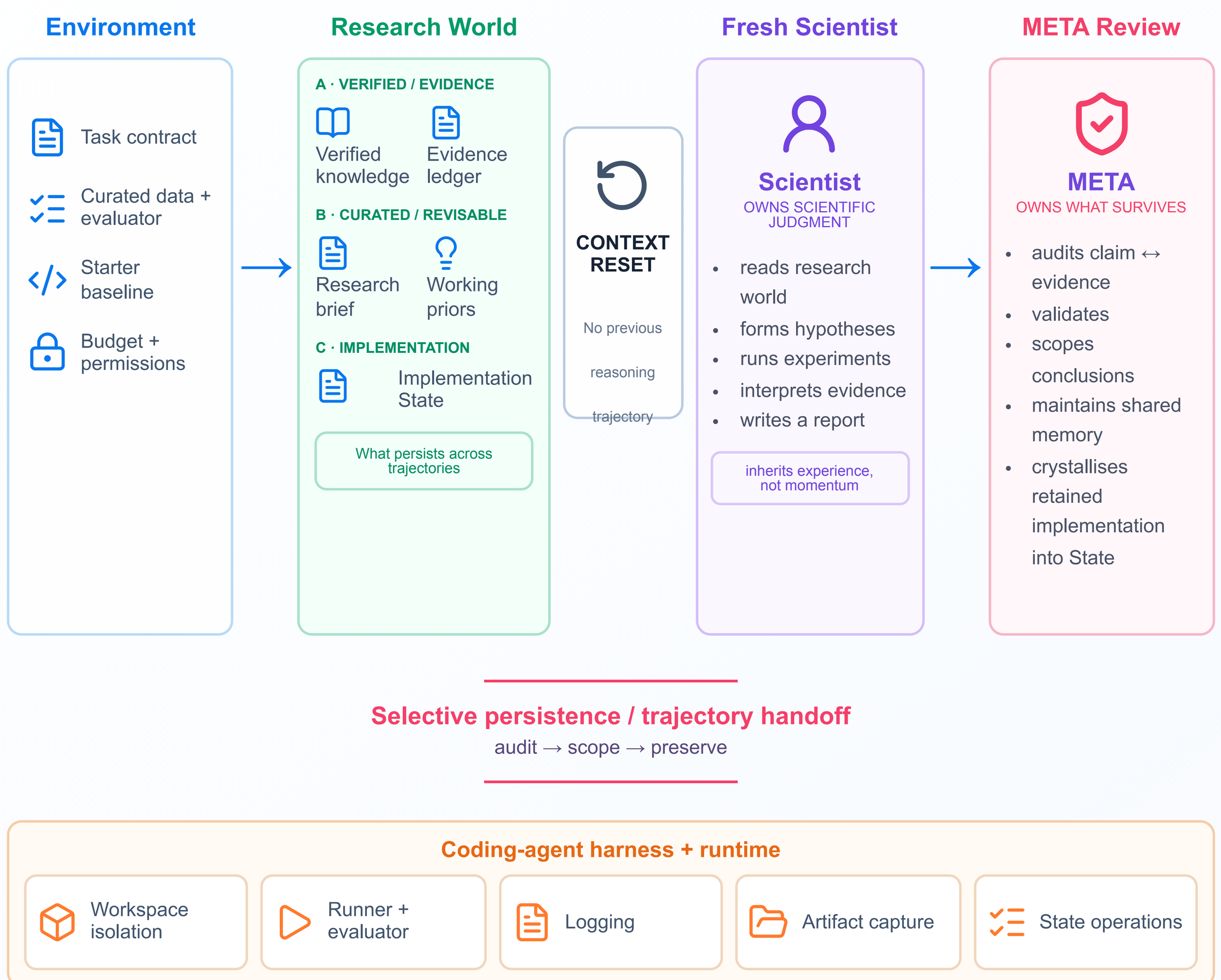
One-edit read/write ratio: 8.03×. Eight-child reuse projection: 4.51×.

02 | SYSTEM

SCIENTIST ↔ META COLLABORATION

A system for sustained research

Give the research world and the researcher different lifetimes.



WORKTREE ISOLATION

Parallel branches explore in separate Git worktrees. Review compares the evidence before an explicit branch promotion.

AUDITED HANDOFF

The next Scientist reads retained evidence, working priors and implementation State. META scopes what survives each trajectory.

03 | USER EXPERIENCE

A research layer over your agent harness.

Works with Codex · Claude Code · OpenCode · Antigravity · Trae

`task.md`

What to solve

Objectives · constraints · evaluation

Define the task → Delegate research

`PERSONAL.md`

How you want it solved

Preferences · research style · priorities

Review evidence

↔ Coding agent — EXAMPLE INPUT

```
> Use the research-agent skill for task.md.
  Follow PERSONAL.md. Run autonomously.
  Let me review the results and evidence.
```

01	02	03	04	05
Define task	Single step	Run research	Explore in parallel	Review evidence

04 | EVALUATION

Our agent can carry a research task to completion.

The model got better.

0.6016000 Official FM baseline → **0.6059363** SciOdyssey · +0.0043363 Primary

Metric	Official FM	SciOdyssey	Delta
Primary	**0.6016000**	**0.6059363**	**+0.0043363**
GAUC	**0.6674000**	**0.6728421**	**+0.0054421**
nDCG@5	**0.5357000**	**0.5390304**	**+0.0033304**

Primary = mean(GAUC, nDCG@5). Final recipe: 46 categorical feature interactions · 8-seed FM · rank 16 · NumPy / CPU.

The search stayed productive.

CYCLE 1	CYCLE 2	CYCLE 3	CYCLE 4
0.6016310	**0.6040901**	**0.6044289**	**0.6059363**
Valid baseline 15-field FM	Feature expansion 38 fields · 5 seeds	Variance reduction 8-seed ensemble	Retained recipe 46 interactions

4 autonomous cycles · **13** named experiments · **0** manual scientific interventions*

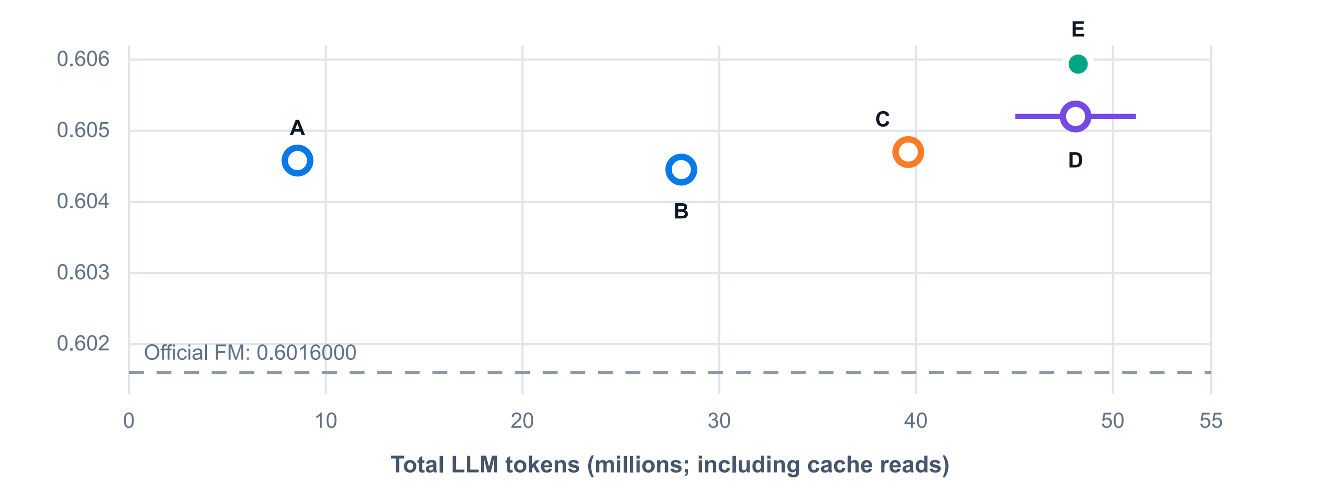
7 full evaluations · 124,909 validation rows · 22,377 users.
170,588 submission rows passed the Starter Kit alignment check.

* After launch, in the retained submission record.

Compare scores alongside their evidence.

KuaiRand-Pure · public validation · one recorded run per condition

Score versus measured LLM-token investment



Recorded agent system	Tokens	Primary
A · Antigravity / gemini-3.7-flash	8.565M	0.6045803
B · Codex / gpt-5.6-luna	28.070M	0.6044533
C · Heterogeneous / gemini-3.7-flash	39.607M	≈0.6047
D · Gemini-only RA / Cycle 2, E008	45.044–51.174M	0.6052
E · SciOdyssey / verified submission	**48.240M**	**0.6059363**

Codex: recorded best 0.6044533 (earlier provisional ≈0.6046). Gemini: gemini-3.7-flash. Cycle-2 range is a token-accounting bound, not uncertainty in score. Models and budgets differ; this comparison does not isolate reset or META effects.

The agent recovered from failure.

Self-healing: In Cycle 1, the Scientist patched numpy.float32 serialization and completed the rerun without human intervention.

META review: In one experiment, sampled training users with unfiltered validation distorted the BPR/BCE ranking. META caught the mismatch and blocked State promotion.

05 | INSIGHTS

01 · MODEL DIVERSITY

Deepen an idea. Open new directions.

In the recorded work, GPT 5.6 Luna refined promising details; Gemini 3.7 Flash explored alternative approaches. The retained Scientist sequence alternated GPT → Gemini → GPT → Gemini, with shared memory maintained by Gemini META.

Gemini-only: **0.6052** · GPT + Gemini: **0.6059363**

Different recipes and stopping boundaries; model rotation was not isolated in an ablation.

02 · PARALLEL RESEARCH

Explore in parallel. Build on the findings.

The reviewer selected branch B (≈0.6048) for further work while retaining modeling ideas from lower-scoring branch C (≈0.6039). A fresh Scientist developed B using C's findings: pairwise ranking, higher rank and rank-normalized ensembling.

Synthesis: **0.6055536** · Public-validation Primary

A lower-scoring branch can still contribute an idea. This reconstructed parallel record is separate from the final submission and does not isolate the effect of parallelism.

03 · ENGINEERING PRACTICE

A clean repo. A continuous agent.

Give every agent a branch. Independent changes keep conflicts local and the main branch clean. **Leave the next move in GitHub.** An issue or prompt can trigger a GitHub Action; review the resulting commit before merging.

Issue / prompt → GitHub Action → Commit → Review

Recommended workflow for unattended exploration, not a measured capability of the retained research run.

RETAINED RUN · 48.240M total tokens · 4.020M non-cache · 0 GPU-hours

References

Repository README; docs/ARCHITECTURE.md; docs/FINAL_REPORT.md §3.5; public submission package and telemetry.

Contact Information

SciOdyssey Team

github.com/zc6600/research-agent-kuairand ↗

Project Website & Demo

sciodyssey.zc6600.wiki ↗

Scan to explore the live wiki, interactive report & artifacts

Resources: ~$10 project subscription-equivalent estimate · GPT Plus + Gemini Pro, 7 days · 1 MacBook M2 · 0 GPU-hours. Includes development and experiments, not per-run cost.